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DETAIL SPECIFICATION

ANTIOXIDANT, 4,4′-METHYLENEBIS(2,6-DI-TERT-BUTYLPHENOL)

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a low volatile, temperature stable antioxidant, 4,4′-methylenebis(2,6-di-tert-butylphenol) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E203 - Standard Test Method for Water Using Volumetric Karl Fischer Titration

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Composition and purity. The 4,4′-methylenebis(2,6-di-tert-butylphenol) shall be a dry crystalline solid at room temperature and shall be of 98 percent minimum purity when tested as specified in 4.5.1.

3.3 Melting point. The melting point shall be not less than 154.0 degrees Celsius (°C) nor greater than 155.2 °C when tested as specified in 4.5.2.

3.4 Benzene insoluble. The benzene insolubles shall be not greater than 0.2 percent when tested as specified in 4.5.3.

3.5 Water content. The water content shall be a maximum of 0.05 percent when tested as specified in 4.5.4.

3.6 Workmanship. The 4,4′-methylenebis(2,6-di-tert-butylphenol) shall be uniform in quality and free from materials that could adversely affect its use when examined as specified in 4.5.5.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows (table I lists the applicable tests required for each particular level of inspection):

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

TABLE I. Applicable tests for inspection. 1/

Test	First article inspection	Conformance inspection	Requirement
Purity (4.5.1)	1	1	3.2
Melting point (4.5.2)	1	1	3.3
Benzene insolubles (4.5.3)	1	1	3.4
Water content (4.5.4)	1	1	3.5
Workmanship (4.5.5)	1	1	3.6

NOTE:

1/ The numbers indicated are the number of unique lots (see 4.4.1.1) to be produced for the applicable testing required.

4.2 Inspection conditions. Unless otherwise specified (see 4.5), test conditions shall be room ambient conditions.

4.3 First article inspection. When required (see 3.1), the first article inspection shall include the tests specified in table I. The sample size shall be as specified (see 6.2). Failure of the first article sample to pass this inspection shall be cause for rejection of the first article sample.

4.4 Conformance inspection. Conformance inspection shall include the tests specified in table I.

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4.4.1 Sampling.

4.4.1.1 Lot. For the purpose of sampling, a lot shall consist of material of a single form manufactured under one set of conditions from one production batch and offered for delivery at one time.

4.4.1.2 Sampling for workmanship. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-1916, verification level II.

4.4.1.3 Sampling for tests. A representative sample of sufficient material to perform the required tests shall be taken from each lot.

4.5 Test procedures. When tested as specified in 4.5.1 through 4.5.5, the 4,4′-methylenebis(2,6-di-tert-butylphenol) shall meet the requirements of 3.2 through 3.6. All chemicals and reagents used in the test methods shall be reagent grade. If the sample fails to pass one or more tests, the lot represented shall be rejected.

4.5.1 Purity. The equipment and procedure to determine the purity shall be as follows.

4.5.1.1 Equipment. The equipment shall be as follows:

a. Beckman DK2 recording spectrophotometer or equivalent (see 6.6).

b. Absolute ethyl alcohol.

c. The usual laboratory equipment for weighing and diluting the samples.

4.5.1.2 Procedure. The procedure shall be as follows:

a. Weigh 400±1 milligrams of the antioxidant sample into a 500-milliliter (mL) volumetric flask and fill to the 500-mL mark with ethyl alcohol. Allow the sample to dissolve in the alcohol.

b. Dilute the sample 1:10 by pipetting 25 mL into a 250-mL volumetric flask and diluting to volume with alcohol.

c. Prepare a series of standards from the sample from step “b” as follows:

5 mL aliquot to 50 mL with alcohol
10 mL aliquot to 50 mL with alcohol
15 mL aliquot to 50 mL with alcohol
20 mL aliquot to 50 mL with alcohol
30 mL aliquot to 50 mL with alcohol
40 mL aliquot to 50 mL with alcohol
sample with no dilution

d. Place ethanol in matched 1-centimeter cells and set the instrument from 0 percent transmittance at 277.5 millimicrons (mu) in accordance with the manufacturer’s instructions.

e. Scan each of the samples from step “c”, and determine its absorbance at 277.5 mu by using the following formula:

$$A = \log \frac{100}{T}$$

Where:

A = absorbance
T = percent transmittance

f. Plot the absorbance values from step “e” as ordinates and concentrations from step “c” as abscissa on graph paper and draw a straight line through the points using the origin as one point.

g. Read the absorbance at the highest concentration. This is obtained from the point of intersection of the line at the most concentrated sample.

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h. Calculate as follows:

$$\% \text{ purity} = \frac{A}{3567 \times 1.88 \times 10^{-6}}$$

Where:

A = absorbance from step "g"

3567 = molar absorbance index determined from a pure sample of 4,4′-methylenebis(2,6-di-tert-butylphenol)

1.88 = concentration factor in moles per liter

4.5.2 Melting point. The equipment and procedure to determine the melting point shall be as follows.

4.5.2.1 Equipment. The equipment shall be as follows:

a. A certified thermometer graduated in 0.2 °C in the 150 to 155 °C range.

b. A second short thermometer to monitor the oil bath temperature.

c. An oil bath. (A 250-mL metal or wrapped glass beaker with mineral oil and an overhead stirrer is satisfactory.)

d. A heater in series with a variable transformer, such as a Variac or equivalent, to control heat input to the bath.

e. Capillary melting point tubes.

4.5.2.2 Procedure. The procedure shall be as follows:

a. Place the bath on the heater and, with the short thermometer in the bath and with continuous stirring, stabilize the temperature at or near 150 °C so that there is no rapid rise or fall in temperature over a 2- or 3-minute period.

b. Place the antioxidant in a capillary melting point tube to a depth of about 5 millimeters and fasten the tube to the thermometer with a collar cut from a piece of rubber tubing of an appropriate size.

c. Lower the thermometer into the bath at about 150 °C and slowly increase the heat to give a temperature rise of about 1 °C per minute.

d. Read the temperature at which the sample is completely melted. This shall be not less than 154 °C nor greater than 155.2 °C.

e. If the melting point is not within the required range, repeat steps "a" through "d" until a melting point has been determined with certainty.

f. Make a stem temperature correction if the thermometer used has not been calibrated for the emergent stem. The melting point shall be the average of two determinations.

4.5.3 Benzene insoluble material. The procedure to determine the benzene insoluble material shall be as follows.

4.5.3.1 Procedure. Take a sample of material whose weight is between 9.500 and 10.500 grams. Dissolve the sample in 102.5 mL of benzene. Filter the mixture through a fluted filter paper that has been dried to a constant weight in a steam-heated oven. Wash the residue with 50 mL of benzene. Dry the fluted filter paper and residue to a constant weight in a steam-heated oven. Calculate the percent benzene insoluble material by using the following formula:

$$\text{Percent} = \frac{R}{W} \times 100$$

Where:

R = weight of residue in grams to the nearest milligram

W = weight of sample in grams to the nearest milligram

NOTE: All operations using benzene shall be performed in a hood.

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4.5.4 Water content. The water content shall be determined in accordance with ASTM E203.

4.5.5 Workmanship. The samples taken in accordance with 4.4.1.2 shall be visually examined for contamination with extraneous material and uniformity using daylight illumination.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 4.4′-methylenebis(2,6-di-tert-butylphenol) covered by this specification is intended for use in the manufacture of liner material for torpedo warheads and as a component of plastic-bonded explosive formulations. Existing stock of 4.4′-methylenebis(2,6-di-tert-butylphenol) is acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity desired.

c. When first article inspection is required (see 3.1).

d. First article sample size, if required (see 4.3 and 6.3).

e. Packaging requirements (see 5.1).

f. Safety data sheet (SDS) requirements (see 6.4).

6.3 First article. When first article inspection is required, the contracting officer will provide specific guidance to offerors whether the sample(s) will be a preproduction sample, a first article sample, a first production sample, a sample selected from the first production items, or a standard production sample(s) from the contractor's current inventory (see 3.1), and the size and number of samples to be tested as specified (see 4.3). The contracting officer will also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first article samples. When the 4,4′-methylenebis(2,6-di-tert-butylphenol) is used as a constituent of energetic materials required to comply with AOP-7 and STANAG 4170, the contracting officer will contact the Technical Authority (see 6.5) to determine the number and size of the samples and an acceptable laboratory for evaluation. Invitations for bids will provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation. Generally, a first article inspection, or part of a first article inspection, will be waived only if the following requirements have been met:

a. The manufacturer has made no changes to their process, procedures, or ingredient manufacturers since last qualified.

b. There has been no interruption in the manufacturer's production of 4,4′-methylenebis(2,6-di-tert-butylphenol) exceeding 12 months at the time of contract award.

6.4 SDSs. When required (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

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6.5 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.6 Manufacturer information. The spectrophotometer used for purity testing is available from Beckman Coulter, Inc., CAGE: 6Z677, www.beckman.com, 800-526-3821, 250 South Kraemer Boulevard, Brea, CA 92821-6232; or may be available from equivalent suppliers.

6.7 Subject term (key word) listing.

Energetic ingredient

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – OS

Review activity:
DLA – GS

Preparing activity:
Navy – OS
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